

PRESENTATION ON

GAN-BASED IMAGE DE-NOISING AND MULTI-ANGLE VIEW GENERATION

A Deep Learning Approach for Facial Image Restoration



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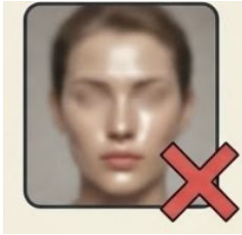
Introduction

⚠ The Challenge



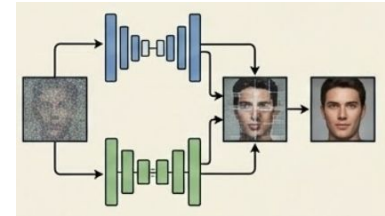
Images captured in uncontrolled environments often suffer from noise. This degrades quality significantly, making them unusable for face recognition tasks.

⊘ Limitations



Traditional filters (Gaussian, Median) and standard CNNs tend to over-smooth images. They remove critical high-frequency details like textures and edges.

💡 Our Solution



We propose a GAN-based framework using U-Net and PatchGAN. This learns to hallucinate realistic details from noisy inputs.

Problem Statement

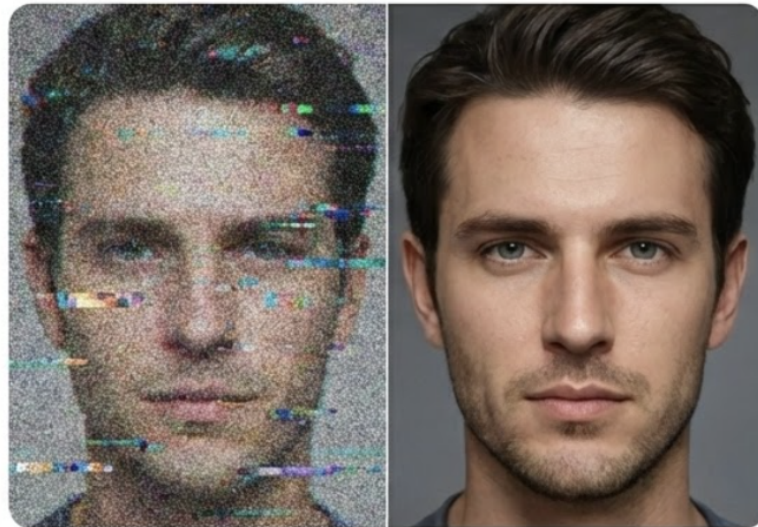
1. Loss of Fine Details

Noise corrupts critical facial features. Standard restoration results in "waxy" or blurred faces that lack identity.

2. Complex Dependencies

Restoring a face requires preserving **Structural Similarity (SSIM)** while ensuring perceptual realism. Simple pixel-loss models fail here.

Noisy vs Clean Male Face Comparison



Research Objectives



Develop GAN Model

Create a GAN to remove noise and restore image quality efficiently.



Multi-Angle Foundation

Implement framework compatible with future multi-angle generation.



Preprocessing Pipeline

Build a robust pipeline for preparing noisy and clean dataset pairs.



Evaluation

Assess performance using PSNR and SSIM metrics.

Literature Review: DnCNN (Zhang et al., 2017)

CORE IDEA

Utilizes deep CNNs with residual learning to predict the "noise map" rather than the clean image.

CONTRIBUTION

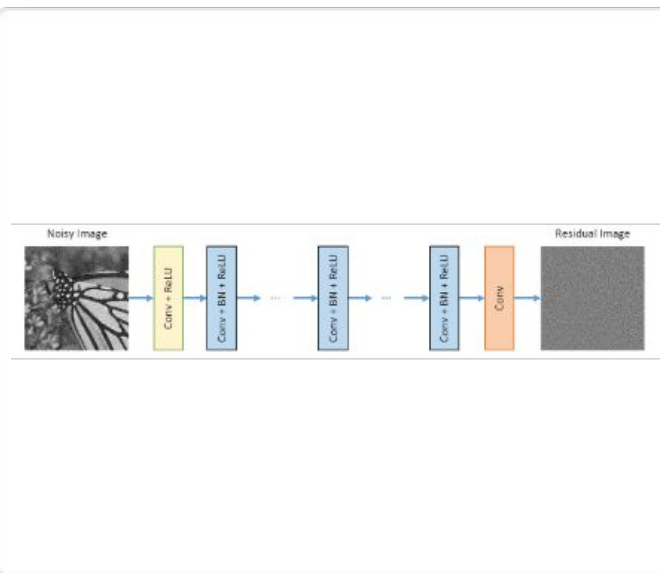
Introduced residual learning to image denoising, significantly speeding up the training process.

MODEL STRUCTURE

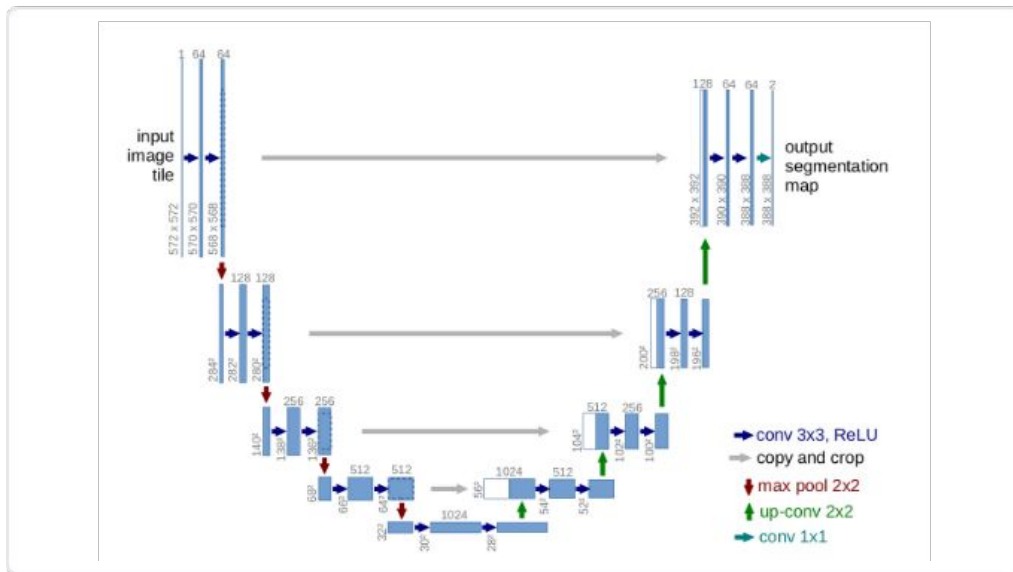
Deep Convolutional Neural Network with Batch Normalization (BN) and ReLU layers.

RESULT

High PSNR but often produces over-smoothed textures, lacking facial details.



Literature Review: U-Net (Ronneberger et al., 2015)



CORE IDEA

Encoder-Decoder architecture that preserves spatial info during downsampling.

CONTRIBUTION

Introduced **Skip Connections** that transfer low-level features directly to the decoder.

MODEL STRUCTURE

Symmetric U-shaped architecture with contracting path (encoder) and expansive path (decoder).

RESULT

Crucial for recovering fine details like eyes and hair in face restoration.

Literature Review: Pix2Pix (Isola et al., 2017)

CORE IDEA

Conditional GANs (cGAN) mapping input pixels to output pixels directly.

CONTRIBUTION

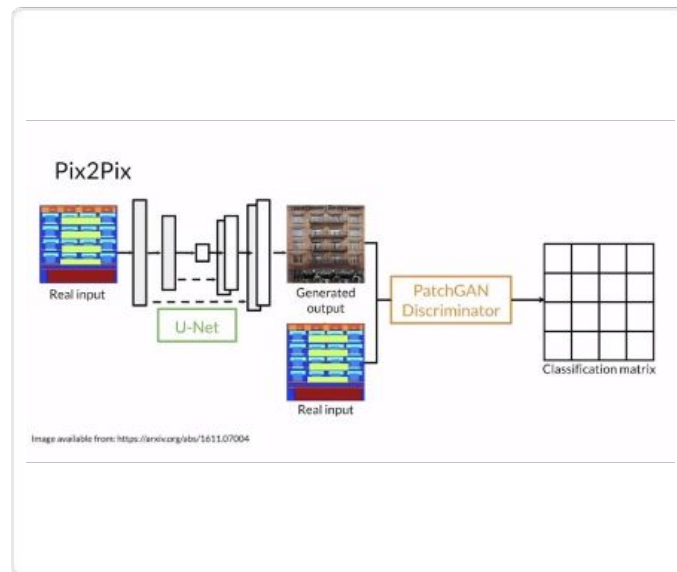
Proposed the **PatchGAN** discriminator.

MODEL STRUCTURE

Generator (U-Net) + Discriminator (PatchGAN) trained with Adversarial Loss.

RESULT

Evaluates realism at the scale of local patches ($N \times N$), enforcing sharp, realistic textures.



Dataset: CelebA

We utilize the **CelebFaces Attributes Dataset (CelebA)**.

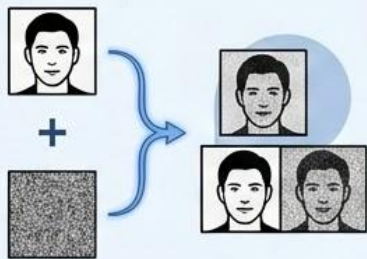
- **Source:** CelebA Dataset – CUHK (Liu et al., ICCV 2015), accessed via Kaggle
- **Scale:** Subset of 2,400 images (2000 Train, 400 Val).
- **Resolution:** Resized to 128×128 pixels.
- **Processing:** Gaussian Noise ($\sigma = 0.10$) added.

Training Images



METHODOLOGY

1. Noise Injection

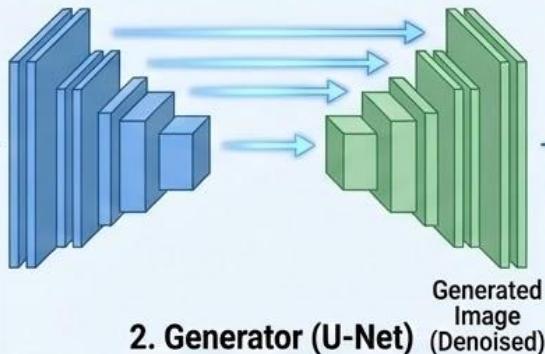


Noisy Input

1. Noise Injection
Clean images are corrupted with Gaussian noise to create paired training data.

Paired Data

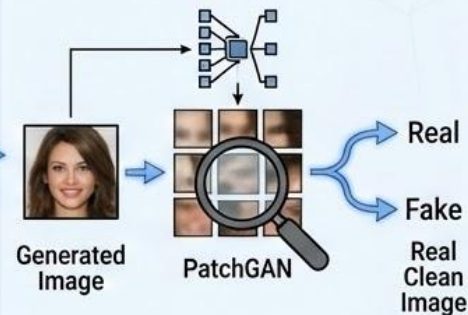
2. Generator (U-Net)



2. Generator (U-Net)
Encoder captures context; Decoder recovers spatial details via skip connections.

Generated Image

3. Discriminator



Generated Image

3. Discriminator

PatchGAN classifies patches as "Real" or "Fake" to enforce texture realism.



4. Multi-Loss

L1 + Perceptual + Adversarial Loss.

Result Metrics

METRIC 1

PSNR

Peak Signal-to-Noise Ratio.
Measures pixel-level fidelity.

METRIC 2

SSIM

Structural Similarity Index.
Measures preservation of structure.

Conclusion

Success

Designed a **GAN-based framework** that aims to outperform traditional filters in preserving details.

Approach

Combining **U-Net** and **PatchGAN** to achieve high perceptual realism and structural accuracy.

Future Work

This denoising module serves as the critical pre-processing foundation for *Multi-Angle View Generation* in the next semester.

References

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- 3 Isola, P., Zhu, J. Y., Zhou, T., & Efros, A. A. (2017). Image-to-Image Translation with Conditional Adversarial Networks. *CVPR 2017*.
- 4 Zhang, K., et al. (2017). Beyond a Gaussian Denoiser: Residual Learning of Deep CNN. *IEEE TIP 2017*.
- 5 Liu, Z., et al. (2015). Deep Learning Face Attributes in the Wild. *ICCV 2015*.
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